Andy Yu

+1 617 383 3149| andy2014.yu@gmail.com | linkedin.com/in/andyyu03

EDUCATION

ETH Zurich
Master of Science: Information Technology & Electrical Engineering — ***2024 – Present***
Bachelor of Science: Information Technology & Electrical Engineering — ***2021 – 2024***

WORK EXPERIENCE

Thales – Secure Communication and Information Systems — **Zurich, CH**
Intern, Systems Engineer and Solution Architect — *May 2026 – Sep 2026*

- Defined and owned system level functions for military communication systems, including functional decomposition, interface definition, requirement allocation, and end to end system behavior
- Integrated hardware, firmware, and software components in collaboration with development teams using C++, Verilog, VHDL, FPGA-based designs including debugging
- Planned and executed system integration, verification and validation activities, including laboratory measurements, field testing, fault isolation and performance characterization
- Developed test procedures and acceptance criteria for system functions, interfaces, timing behavior, RF performance, and hardware software interaction
- Supported customer related technical activities through requirement clarification, system demonstrations, field trials, troubleshooting, and preparation of specifications, interface documents, test reports, and verification evidence

ETH Zurich – Institute for Signal and Information Processing — **Zurich, CH**
Teaching Assistant — *Feb 2025 – Jun 2025*

- Supported laboratory instruction in digital signal processing, communication systems, control circuits, and machine learning, guiding students through theoretical concepts and implementation
- Supervised MATLAB/Simulink and Python exercises covering continuous phase modulation, digital filtering, sampling-rate conversion, channel equalization, error-correcting codes, neural-network regression and clustering algorithms
- Helped students debug simulation models, signal processing code, electronic circuits, and measurement setups while explaining performance metrics such as spectral efficiency, aliasing, decoding error probability and reconstruction accuracy

ETH Zurich – Institute for Integrated Systems — **Zurich, CH**
Lead Teaching Assistant — *Sep 2024 – Feb 2025*

- Supervised undergraduate electronics laboratories covering diode characteristics, rectifier circuits, active filters, BJT and MOSFET amplifier stages, biasing, impedance analysis, voltage gain and frequency response
- Supported circuit simulation and hardware implementation using NI Multisim and NI ELVIS II+ prototyping systems, including oscilloscopes, function generators, digital multimeters, Bode analyzers and I–V analyzers
- Guided students through circuit assembly, component selection, instrument configuration, analytical calculations and systematic debugging of measurement and simulation discrepancies
- Assisted with characterization of semiconductor devices and amplifier circuits through I–V measurements, transient analysis, Bode plots, cutoff frequency estimation, gain measurements, and load-dependence testing

Compal Electronics USA — **Logansport, IN, USA**
Intern, Test Engineering — *Jul 2022 – Sep 2022*

- Developed and validated PCB schematics for fixture electronics interfacing data-acquisition systems with manufacturing equipment, using KiCad for schematic capture and PCB layout
- Applied PCB design principles including high speed and high frequency signal integrity, transmission line behavior, S-parameters, four-port network analysis, impedance matching and component selection

PROJECTS

Probabilistic Transformer based Modeling of GNSS scintillation during geomagnetic storms — ***Sep 2026 – Present***
Supervisors: Dr. Yukitoshi Nishimura (Boston University), Dr. Christoph Studer (ETH Zurich)
*Research conducted under **NASA's SPARTA** project, involving MIT, Boston University, Boston College, Cornell University and University of Colorado Boulder*

- *Build end to end deep learning pipeline using Transformer, GRU and CNN's to predict storm time irregularities*
- Transformed and aggregated billons of raw GNSS data to controlled parquet for optimized storage, training and inference efficiency
- Integrated geomagnetic indices, satellite geometry, spatiotemporal features to model ionospheric disturbances
- Evaluated Performance using regression metrics, extreme event detection and upper tail analysis

Diffusion Models for Physical Layer Communication — ***Sep 2025 – April 2026***
Supervisors: Dr. Lutz Lampe (UBC), Dr. Christoph Studer (ETH Zurich)

- Build diffusion models for time-series prediction of wireless channel coefficients, improving channel prediction accuracy in highly time and frequency selective environments
- Transformation of problem setting into Delay Doppler Angular (DDA) domain
- Designed and tested Massive MIMO signal processing methods for both single- and multi-user systems, leveraging pilot-based channel estimation to enhance prediction robustness in complex fading environments
- Achieved 5–10 dB NMSE improvement vs. classical & data driven baselines

Urban Sound Classification using a 1D CNN on an MCU — ***Sep 2024 – Jan 2025***
Supervisor: Dr. Michele Magno (ETH Zurich)

- Design and deployment of lightweight CNN for live inference using Urban8K dataset achieving accuracy of 85+% on microcontroller
- Optimized model architecture with TinyML principles (pruning, quantization) on resource-constraint platform
- Engineered on device feature extraction pipeline, ensuring efficiency and low latency (1.2ms inference time)

Beamforming Modeling of Large Scaled Phased Arrays — ***Feb 2024 – Jul 2024***
Supervisor: Dr. Hua Wang (ETH Zurich)

- Analyzed impact of phase and amplitude quantization on gain, side lobe level and accuracy, directly informing performance tradeoffs in real world deployment
- Evaluate how reduced hardware affects beamforming robustness and coverage
- Identify array configurations that are optimized for performance, complexity and power consumption
- Built reusable MATLAB framework enabling exploration of array geometries, quantization strategies, supporting scaleable antenna design studies

SKILLS

Languages: German (native); English (fluent); French, Mandarin (conversational)
Programming & Software Engineering: Python, C++, C, MATLAB, REST API development with FastAPI and Flask, Jira
Machine Learning & Big Data: PyTorch, TensorFlow, Keras, Scikit-learn, CUDA, deep learning, distributed data processing with Apache Spark, SQL, MongoDB
Cloud & Infrastructure: Docker, Git, GitHub Actions, CI/CD, MLflow, AWS including EC2, S3, SageMaker, Azure, Database design
Embedded, FPGA & Signal Processing: Verilog, VHDL, microcontrollers, DSP, Assembly
Communications Domain: LTE, 5G NR, 6G, 3GPP, QuaDRiGa, TCP/IP, UDP, QUIC, BGP, forward error correction, information theory, wireless channel modeling